Correlated Multi-armed Bandits

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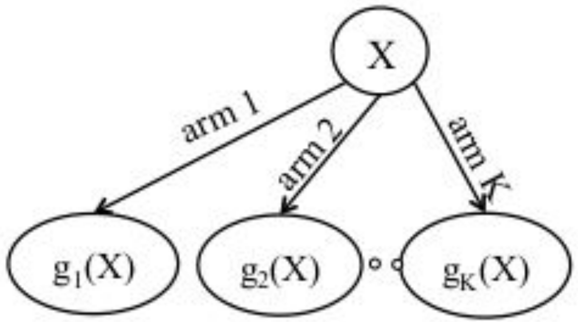
Motivation

- In many practical situations, rewards across arms are not independent
- Pulling one arm of a MAB instance, can also give useful information about the rewards of other arms
- Correlated MAB: Rewards are deterministic functions of a common latent random variable

Problem Definition

- Latent variable X - discrete or continuous
- K - total number of arms, $g_k(x)$ is reward associated with arm k
- Reward $g_{k_t}(x_t)$ where k_t is the arm pulled at time t, and x_t **is an i.i.d realization of X.**
- Assumption: Reward functions are bounded within an interval of size B. (UCB also needs this) i.e. $\max_x g_k(x) - \min_x g_k(x) \leq B \ , \forall k$



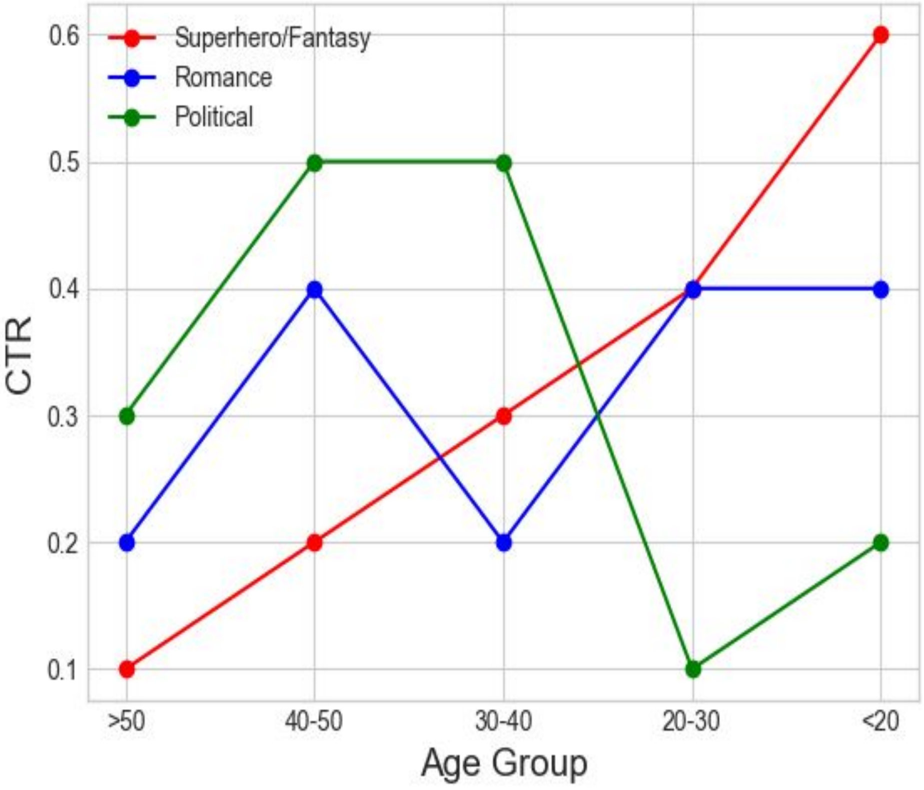
Say 3 new movies are releasing on Friday

The click through rates of different age groups have been estimated(and are known to be true) as shown in the graph. (Controlled Trials)

Need to estimate the distribution of people logging into your app/webpage.

The goal is to maximise the CTR so as to maximise revenue.

Here, X refers to the age group, and g are the different reward functions.



Definitions

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- Pseudo-gap: Of arm l w.r.t arm k $\Delta_{l,k} := \mu_k - \phi_{l,k}$

  - What if pseudo-gap is positive?

  - Define arm l to be competitive w.r.to arm k only if pseudo-gap < 0

C-UCB Algorithm

1. **Input**: Reward Functions $\{g_1 \ldots g_K\}$
2. For each round t:
3. k_{max} := Arm pulled maximum times till t-1
4. Initialize competitive set = [K]
5. For each arm k ($k \neq k_{max}$):
6. If $\hat{\mu}_{k_{max}} > \hat{\phi}_{k,k_{max}}$, remove arm k from competitive set
7. Apply UCB over arms in competitive set
8. Update empirical reward and empirical pseudo-reward

Regret Improvement: $K*\log(t) \longrightarrow C*\log(t)$

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- Max no. of samples, so can accurately identify non-competitive arms
- Can we do better?

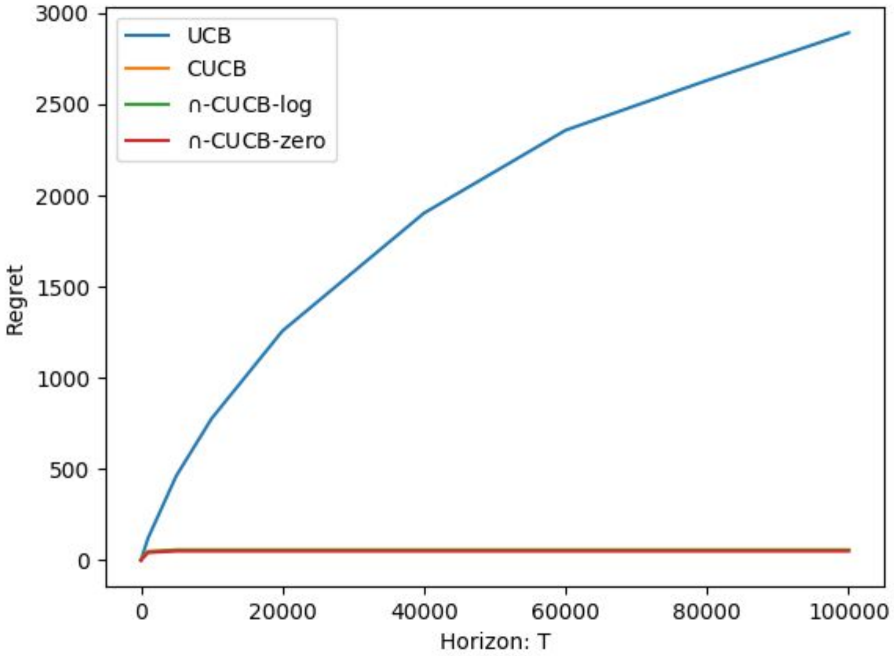
$\cap$-CUCB Algorithm

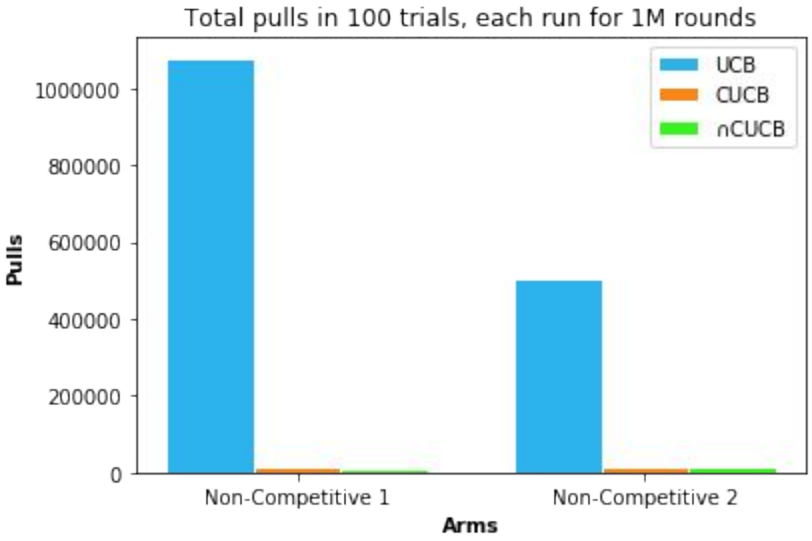
1. **Input**: Reward Functions $\{g_1 \ldots g_K\}$, Cutoff Function C(t)
2. For each round t:
3. Initialize competitive set = [K]
4. For each k' s.t. n(k') > C(t):
5. Current Competitive Set = [K]
6. For each arm k (k != k'):
7. If $\hat{\mu}_{k'} > \hat{\phi}_{k,k'}$, remove arm k from Current Competitive Set
8. Competitive Set = Competitive Set $\cap$ Current Competitive Set
9. Apply UCB over arms in competitive set
10. Update empirical reward and empirical pseudo-reward

Why would the intersection be non-empty?

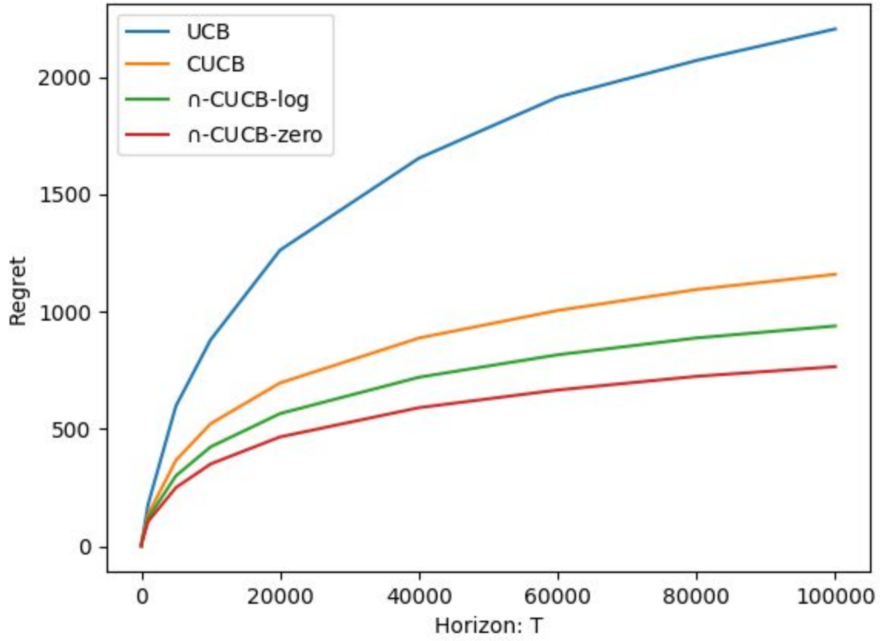
- Trivial to see non-emptiness at t=0. (The entire set is competitive)
- If k_t is picked at time t, only competitive set of k_t is updated.
- Since k_t always belongs to its own competitive set, it stays in the intersection.
- Therefore the arm picked always stays in the intersection and thus the set cannot be empty.

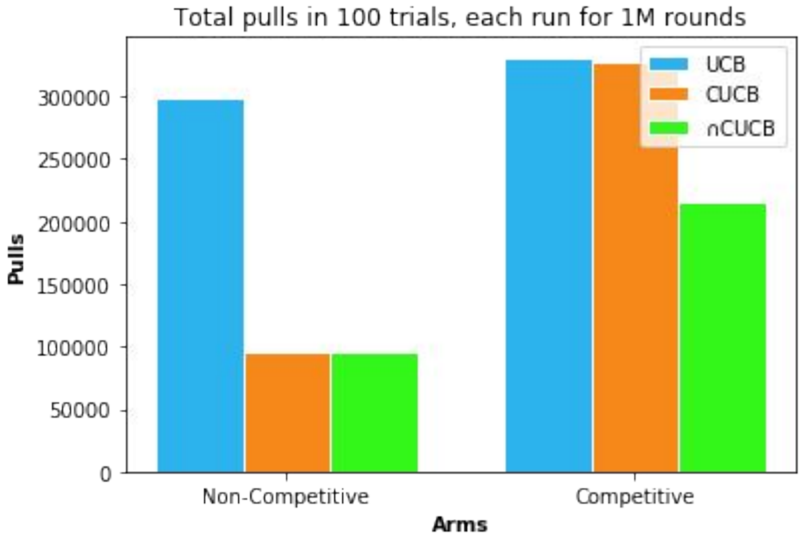
No Competitive Arms



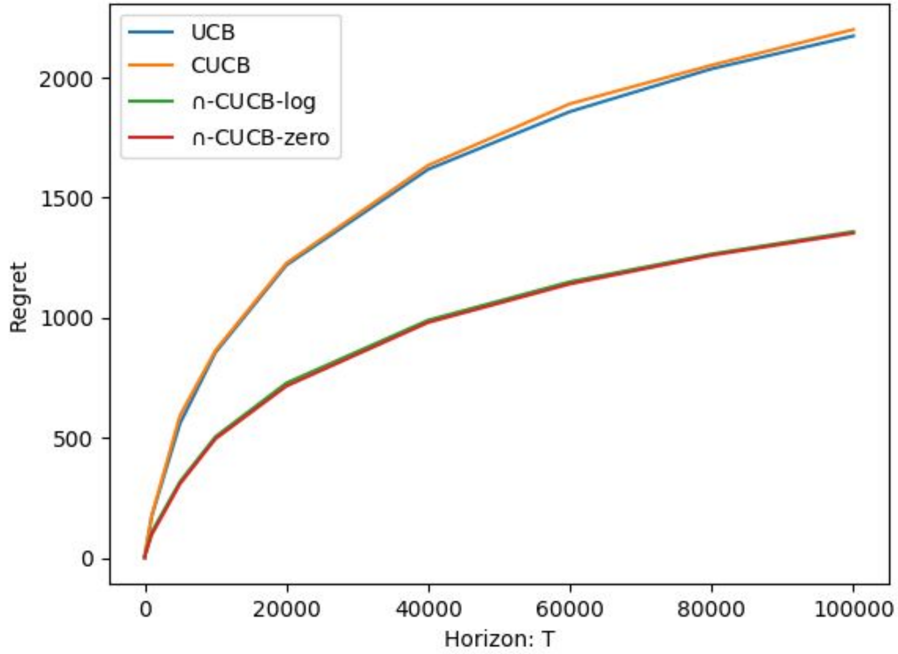


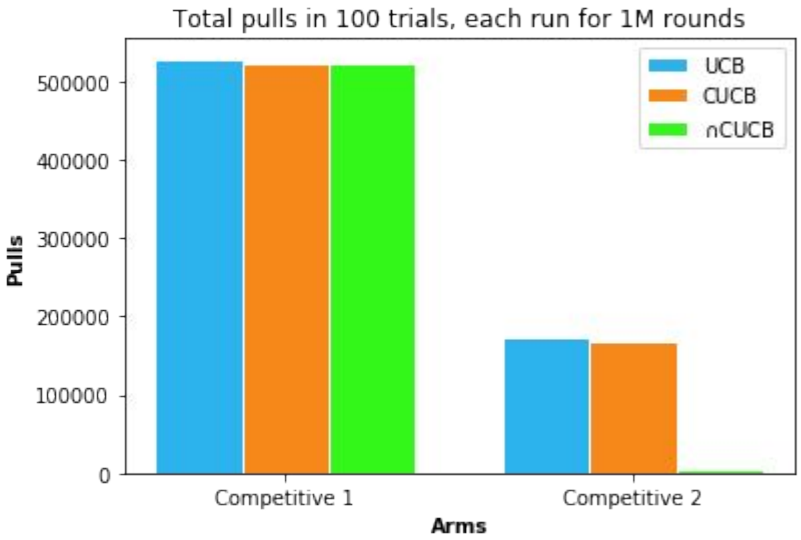
One Competitive Arm





All Competitive Arms





Theoretical Analysis

- Regret Analysis
  - Same order of regret if C(t) increases at least as fast as $4log(t)/k$
  - Empirically C(t) = 0 was best on the attempted test cases.

- What remains to be proven?
  - For C(t) = 0, optimal arm is empirically non-competitive at time t with very small probability (less than t^{-3})
  - Since intersection already includes k_{max}, above is sufficient to prove our algorithm is at least as good as CUCB.

Failed Attempts

- Compute Better Estimate of ϕ (expected pseudo reward)
  - CUCB only used pseudo rewards w.r.t arm pulled max times.
  - Use all pseudo rewards instead for better estimation!
  - Observation: The optimal arm is also removed from the competitive set in a fairly large proportion of trials giving bad (linear!) regrets.
- Linear Cutoffs:
  - Works, but approaches C-UCB very soon.
  - Reason: Pulls of non-optimal arms are of order of log(t). So, after some time only best arm remains with pulls more than cutoff.

References

1. Correlated Multi-armed Bandits with a Latent Random Source - https://arxiv.org/pdf/1808.05904.pdf
2. Exploiting Correlation in Finite-Armed Structured Bandits - https://arxiv.org/pdf/1810.08164.pdf

Thank you!